

ZAFAR ULLAH

AI/ML Engineer | Agentic AI Specialist | Generative AI Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in agentic systems, retrieval-augmented generation (RAG), and production-oriented LLM applications. Experienced in building multi-agent workflows with LangGraph, RAG pipelines using LangChain and vector search (FAISS), and FastAPI-backed AI services integrated with Groq, OpenAI, and Gemini APIs. Applied background spans agriculture, finance, environmental analytics, and conversational AI, with a consistent focus on moving models from experimentation into explainable, deployable systems. Actively developing an industry-facing portfolio of agentic AI, computer vision, and NLP projects while pursuing a B.Sc. in Computer Science with an AI specialization.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript

Machine Learning & Data Science: Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, feature engineering

Deep Learning: TensorFlow, PyTorch, Keras, CNNs, ResNet-50, EfficientNetB0, transfer learning

Generative AI & LLMs: Prompt engineering, LLM application development, Groq API, OpenAI API, Gemini API

Agentic AI: LangChain, LangGraph, multi-agent systems, tool calling, autonomous agents

Natural Language Processing: Text classification, sentiment analysis, conversational AI

Computer Vision: OpenCV, Pillow, image classification, image-quality assessment

Backend & API Development: FastAPI, REST APIs, Pydantic, Uvicorn, Pytest

Databases & Vector Stores: FAISS, ChromaDB, SQLite, vector databases

Frontend Technologies: Streamlit, HTML, CSS

DevOps & Development Tools: Docker, Docker Compose, Git, GitHub, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Machine Learning Intern — Flyrank Pakistan

- Contributing to applied machine learning tasks within an internship team, working across data preparation, model evaluation, and Python-based ML workflows.
- Collaborating with team members on production-oriented AI/ML deliverables using established engineering practices.

Junior AI Engineer — AlgoHub Pakistan

- Developing AI/ML solutions leveraging LangChain, LangGraph, Groq, and FastAPI as part of the engineering team.
- Supporting AlgoHub's internship program by creating structured technical content, including internship handbooks for Machine Learning and RAG & Generative AI cohorts.
- Working across the AI development lifecycle, from experimentation to deployment-oriented implementation.

AI Developer, Generative AI Team — Arch Technology Pakistan

- Worked within a generative AI team on LLM-oriented application development during a focused internship engagement.
- Applied Python and generative AI tooling to contribute to team-based technical deliverables.

Developer Intern — Developer Hub Pakistan

- Contributed to software development tasks as part of a structured internship program, applying core programming and engineering practices.

Generative AI Intern — DecodeLabs Pakistan

- Gained hands-on exposure to generative AI concepts and tooling through an internship focused on applied LLM-based development.

AI/ML Engineer — Independent Research and Projects

- Designed and built end-to-end AI/ML systems spanning agentic AI, computer vision, NLP, and generative AI, applying LangChain, LangGraph, FastAPI, PyTorch, and FAISS/ChromaDB.
- Developed multi-agent architectures for domain-specific applications including agriculture, finance, and conversational commerce.
- Maintained a growing portfolio of independent AI/ML repositories documenting model development, evaluation, and deployment work.

SELECTED AI/ML PROJECTS

Fashion AI Studio

github.com/Zafar488/Fashion-AI-Studio

End-to-end, multi-agent fashion product classification platform.

- Developed a fashion product image-classification system supporting 31 fashion categories using TensorFlow, Keras, and EfficientNetB0.
- Built a multi-agent decision pipeline covering image-quality assessment, classification, confidence calibration, prediction auditing, human-review routing, and feedback collection.
- Delivered both a Streamlit interface and a production-oriented FastAPI REST API, using OpenCV, Pillow, Pydantic, Pytest, and Uvicorn; integrated Groq for explainable AI summaries while keeping the computer-vision model as the source of truth; used SQLite for classification history, feedback persistence, and retraining-queue management.

FinGuard AI

github.com/Zafar488

Fraud detection system built on an agentic workflow architecture.

- Designed a fraud detection pipeline using LangGraph to orchestrate multi-step decisioning logic.
- Integrated machine learning components with an agent-based workflow for transaction analysis.

AgriMind Pakistan

github.com/Zafar488

Bilingual agricultural advisory system with an 11-agent LangGraph architecture.

- Built a production-oriented advisory platform combining Groq LLM (Llama 3.3 70B), ResNet-50-based crop disease detection, and FAISS-backed RAG.
- Delivered a FastAPI backend and a six-page Streamlit frontend supporting English and Urdu, deployed with Docker across a phased development process.

SupplyMind

github.com/Zafar488

Demand forecasting and autonomous reordering agent.

- Developed a forecasting pipeline paired with an autonomous agent for inventory reordering decisions.
- Applied agentic workflow design to connect forecasting output with automated business logic.

ZK Restaurant Agents

github.com/Zafar488

WhatsApp-style conversational ordering assistant.

- Built a Streamlit-based chatbot using LangChain and Groq with tool-dispatch logic driven by structured response signals.
- Iteratively refined conversational UI and LLM output quality across multiple development cycles.

SmogNet

github.com/Zafar488

Multi-model air quality forecasting pipeline for Pakistan — Second Place, national datathon.

- Trained models on a dataset of more than 264,000 rows, achieving 99.79% accuracy with a Random Forest classifier.
- Implemented anomaly detection, rule-based pollution-source classification, and a three-tier air quality alert system.

Additional AI/ML Projects

GitHub: github.com/Zafar488

- **Multi-Modal Medical Diagnosis** — AI-assisted diagnostic support system integrating multiple medical data modalities using deep learning models (PyTorch, computer vision).
- **GlobalCart AI Support Engine** — Conversational AI support engine for e-commerce customer queries and order assistance, built with LangChain and FastAPI.
- **AI Law Assistant** — Retrieval-augmented generation assistant for answering legal queries using LangChain and vector search.
- **AI Crop Recommendation Agent** — Agentic system providing personalized crop recommendations from agricultural input data using LangGraph.
- **RAG Knowledge Analyst** — RAG application for analyzing and answering questions over a document knowledge base using LangChain and FAISS.
- **Multimodal Content Engine for Automatic Reel Generation** — Pipeline automating short-form video reel creation from multimodal inputs.

- **Mental Health Support Agent** — Conversational AI agent offering supportive mental health conversations and resource guidance.
- **Text-to-Math Problem Solver** — LLM-based system for interpreting and solving math problems expressed in natural language.
- **RAG Documents Q&A** — Retrieval-augmented generation application for question answering over uploaded documents.
- **Superstore Sales Analytics and Forecasting** — Data analytics and forecasting pipeline for retail sales trend analysis using Pandas and time-series modeling.

EDUCATION

Bachelor of Science in Computer Science — Artificial Intelligence Specialization

University of Engineering and Technology, Mardan | CGPA: 3.56/4.00 (In Progress)

ACHIEVEMENTS

- Second Place — SmogNet Datathon, national competition focused on air-quality forecasting.
- Developed production-oriented AI systems across agriculture, finance, environmental analytics, and conversational AI domains.